

1.1 Introduction to the Course

Dr Peter Kovalsky



What this lecture covers

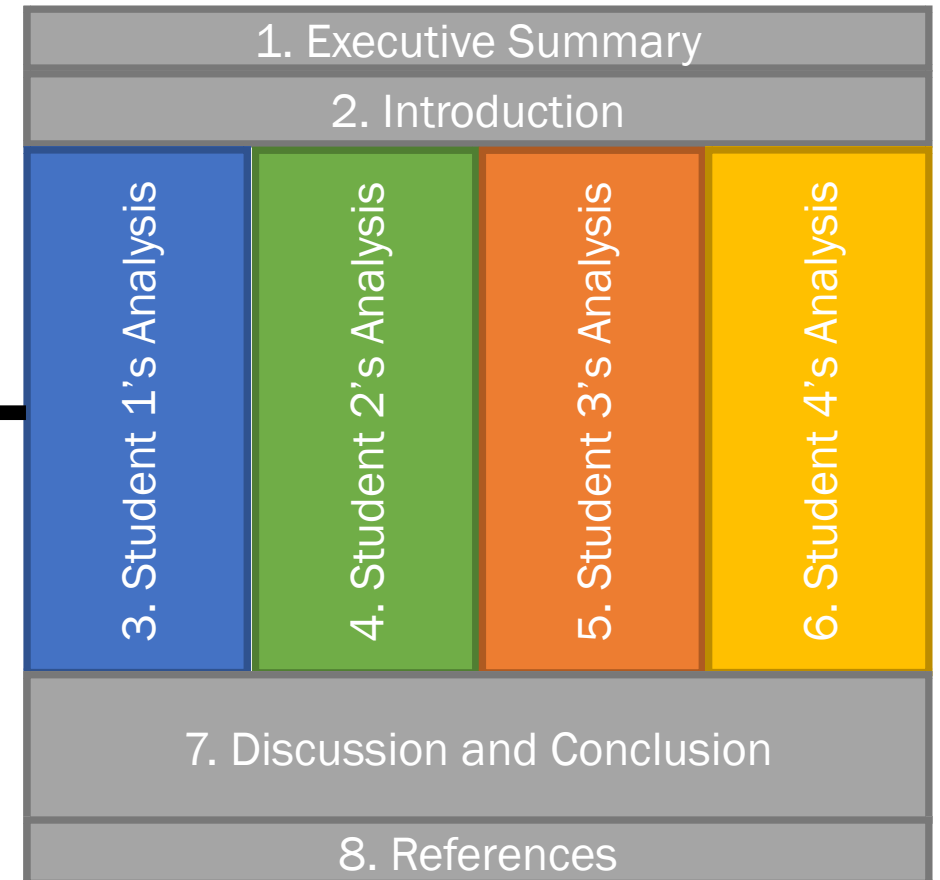
- The course topics, structure and assessment
- Why we have this course
- Engineers and environmental failures of the past

Topics Covered in ENGEN370

1. Design for Environment (3 weeks)
 - *Dr Peter Kovalsky*
2. Life Cycle Assessment (4 weeks)
 - *Dr Mark Lay*
3. Impact Assessment (3 weeks)
 - *Dr Graeme Glasgow*
 - *Dr Mahonri Owen*
4. Risk and Resilience Analysis (2 weeks)
 - *Dr Megan Boston*

Individualised Group Assessment

1. Project Brief – All Group
2. Design for Environment
3. Life Cycle Assessment
4. Environmental Impact Assessment
5. Risk and Resilience Analysis
6. Peer Assessment
7. Short Quizzes



Lab and Group Selection

- You choose your group: four members from at least two different engineering disciplines, all available for the same lab
- Selection is completed via Moodle, due end of the Week 1
- Project Managers (from ENGEN570) will be assigned
- Groups should work together often (at least during the lab sessions) and meet with their Project Manager weekly (outside the lab sessions)

Moodle Page Structure

- Each week a new section topic will appear
- The section will provide
 - *The resources for the week*
 - *Instructions on what you need to do*
 - *The quiz for the week*
- Here's the example for Week 1:

1.1 Design for Environment: Sustainable Management & Design

This week we are going to review the concepts of sustainability, sustainable management, and ethical considerations with respect to the environment, leading into explaining the principles of design for environment.

Your tasks for this week are:

1. Familiarise yourself with the Moodle page, make a lab session selection, and choose a group and lab session
2. Attend the in-person lectures to get a big picture on where this course sits in your degree, how we assess you, and start a discussion on sustainable management and design for environment.
3. Search using Google and contribute to the Environmental failures database by writing a post (about 250 words + an image, if possible) about a significant environmental failure that, ideally, your engineering discipline contributed towards
4. After making your post, complete Quiz 1

 Lecture notes 1

Mark as done

 Environmental failures database

Opened: Monday,

View **Make entries: 1**

 Quiz 1

Opens: Friday,

Closes: Monday,

Receive a grade

Restricted Not available unless: The activity **Environmental failures database** is marked complete



Short Quizzes



Weekly quizzes, open after Friday's lecture, closes Monday midday



Time-limited, normally 15 to 20 minutes



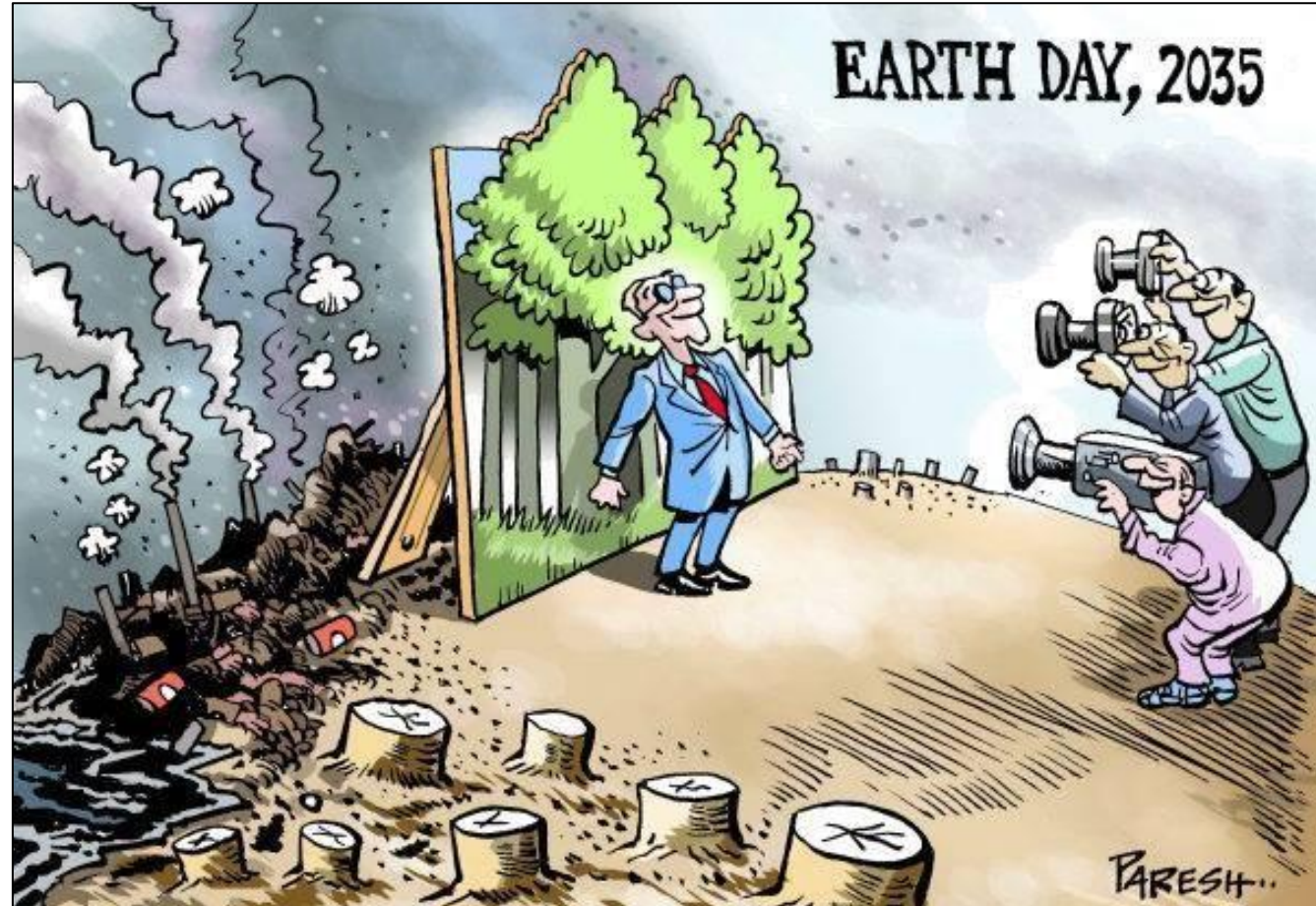
Easy marks if you attend and review the lectures



No groups, no extensions, but exclusion possible

Why have this course?

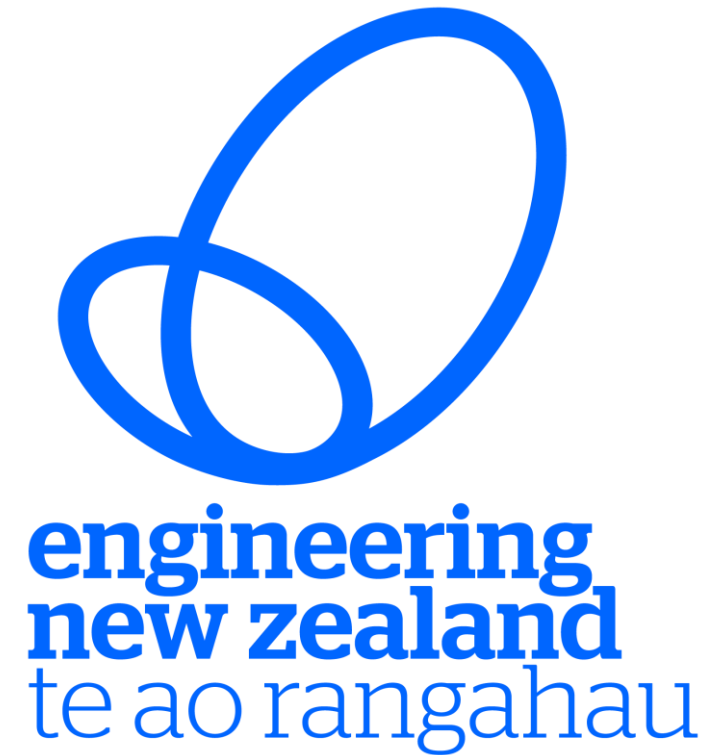
- Globally we face massive, dynamic environmental challenges
- All engineering disciplines are needed to find solutions
- ENZ ethics and Chartership require engineers to care for the environment
- Learnings from past failures are critical



Professional Engineer Chartership: Requirements

To become a Chartered Professional Engineer, you must:

- *Have a Washington Accord-accredited qualification, B.E. Honours*
- *Demonstrate competence*
- *Commit to the CPEng **Code of Ethical Conduct (8 statements)***
- *Be reassessed at least every 6 years*



ENZ: Code of Ethical Conduct

The Code of Ethical Conduct outlines the high standards of ethical behaviour expected of all members of Engineering New Zealand.

1. Take reasonable steps to safeguard health and safety
2. Have regard to effects on environment
3. Report adverse consequences
4. Act competently
5. Behave appropriately
6. Inform others of consequences of not following advice
7. Maintain confidentiality
8. Report breach of Code

2. Have regard to effects on environment

You must, in the course of your engineering activities,

- a) have regard to reasonably foreseeable effects on the environment from those activities
- b) have regard to the need for sustainable management of the environment

What does environment mean:

- a) *ecosystems and their constituent parts, including people and communities; and*
- b) *all natural resources and physical (man-made) resources.*



3. Report adverse consequences

- If you have reasonable grounds to believe that an engineering matter has, or could have, adverse consequences you must bring the matter to the notice of the relevant regulatory body unless, having made inquiries, you are satisfied on reasonable grounds that the matter is being dealt with through an appropriate process or in an appropriate manner.
- *Adverse consequences* means significant damage, or an unacceptable likelihood of significant damage, to the environment

4. Inform others of consequences of not following advice

- If you become aware that your professional advice may not be followed, and consider that a failure to observe that advice may have adverse consequences, **you must** inform the recipient of the advice of those adverse consequences.



Washington Accord – B.E. Outcomes

- WA3: Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
- WA7: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts



Design for the Environment

What is the scope?

- Not just limited to product design.
- It can involve change of process.

A recap of ENGEN 270

Actions that affect the environment can usually be framed within a business context. So what did we learn in Engineering and Business

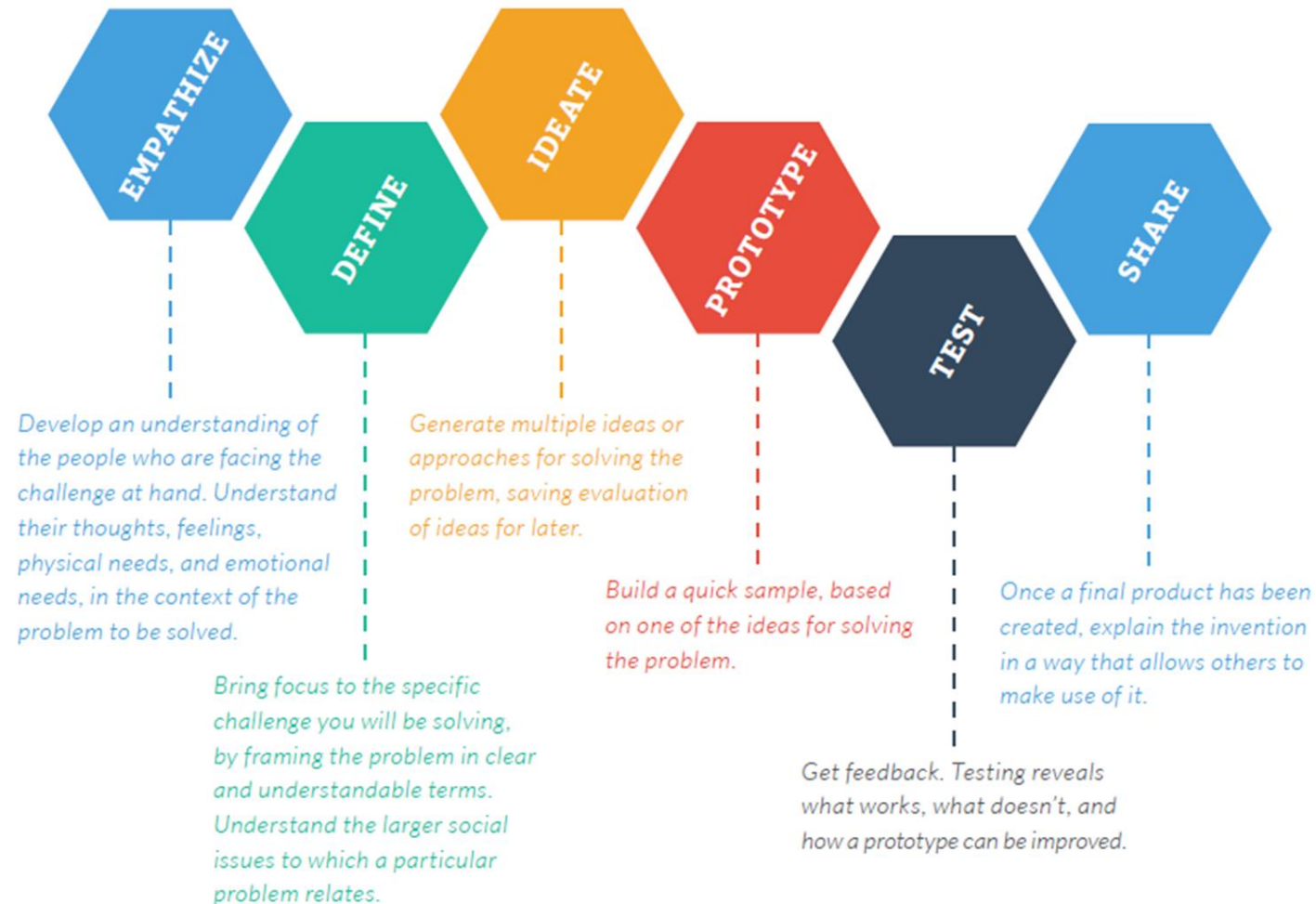
- We learned about lean canvas
- We learned about design thinking
- We learned about project management
- We learned about business finances and return on investment (ROI)

A recap of ENGEN 271/272

Over summer, most would have completed their first work placement.

- Some may have had the opportunity to work within an Environmental Management System like ISO14001. Some may have been exposed to ISO9001.
- Many would have attempted to understand the business by creating a value chain diagram.
- Some may have worked on environmentally themed projects.

Recall Doblin's ten types of innovation and the ideation process



Step 3: Ideate

Profit Model

Premium

Price at a higher margin than competitors, usually for a superior product, offering, experience, service or brand.

Cost Leadership

Keep variable costs low and sell high volumes at low prices.

Scaled Transactions

Maximize margins by pursuing high volume, large scale transactions when unit costs are relatively fixed.

Microtransactions

Sell many items for as little as a dollar—or even only one cent—to drive impulse purchases at volume.

Forced Scarcity

Limit the supply of offerings available, by quantity, time frame or access, to drive up demand and/or prices.

Subscription

Create predictable cash flows by charging customers up front (a one time or recurring fee) to have access to the product/service over time.

Membership

Charge a time-based payment to permit access to locations, offerings, or services that non-members don't have.

Installed Base

Offer a "core" product for slim margins (or even a loss) to drive demand and loyalty; then realize profit on additional products and services.

Switchboard

Connect multiple sellers with multiple buyers; the more buyers and sellers who join, the more valuable the switchboard.

Auction

Allow a market—and its users—to set the price for goods and services.

User-Defined

Invite customers to set a price they wish to pay.

Freemium

Offer basic services for free, while charging a premium for advanced or special features.

Flexible Pricing

Vary prices for an offering based on demand.

Float

Receive payment prior to building the offering—and use the cash to earn interest prior to making margins.

Financing

Capture revenue not directly from the sale of a product, but from structured payment plans and after-sale interest.

Ad-Supported

Provide content/services for free to one party while selling listeners, viewers or "eyeballs" to another party.

Licensing

Grant permission to some other group or individual to use your offering in a defined way for a specified payment.

Metered Use

Allow customers to pay for only what they use.

Bundled Pricing

Sell in a single transaction two or more items that could be sold as standalone offerings.

Disaggregate Pricing

Allow customers to buy exactly—and only—what they want.

Risk Sharing

Waive standard fees/costs if certain metrics aren't achieved, but receive outside gains when they are.

Network

Merger/Acquisition

Combine two or more entities to gain access to capabilities and assets.

Consolidation

Acquire multiple companies in the same market or complementary markets.

Open Innovation

Obtain access to processes or patents from other companies to leverage, extend, and build on expertise and/or do the same with internal IP and processes.

Secondary Markets

Connect waste streams, by-products, or other alternative offerings to those who want them.

Supply Chain Integration

Coordinate and integrate information and/or processes across a company or functions of the supply chain.

Complementary Partnering

Leverage assets by sharing them with companies that serve similar markets but offer different products and services.

Alliances

Share risks and revenues to jointly improve individual competitive advantage.

Franchising

License business principles, processes, and brand to paying partners.

Coopetition

Join forces with someone who would normally be your competitor to achieve a common goal.

Collaboration

Partner with others for mutual benefit.

Structure

Organizational Design

Make form follow function and align infrastructure with core qualities and business processes.

Incentive Systems

Offer rewards (financial or non-financial) to provide motivation for a particular course of action.

IT Integration

Integrate technology resources and applications.

Competency Center

Cluster resources, practices and expertise into support centers that increase efficiency and effectiveness across the broader organization.

Outsourcing

Assign responsibility for developing or maintaining a system to a vendor.

Corporate University

Provide job-specific or company-specific training for managers.

Decentralized Management

Distribute decision-making governance closer to the customer or other key business interfaces.

Knowledge Management

Share relevant information internally to reduce redundancy and improve job performance.

Asset Standardization

Reduce operating costs and increase connectivity and modularity by standardizing your assets.

Process

Process Standardization

Use common products, processes, procedures, and policies to reduce complexity, costs, and errors.

Localization

Adapt an offering, process, or experience to target a culture or region.

Process Efficiency

Create or produce more while using fewer resources—measured in materials, energy consumption or time.

Flexible Manufacturing

Use a production system that can rapidly react to changes and still operate efficiently.

Process Automation

Apply tools and infrastructure to manage routine activities in order to free up employees.

Crowdsourcing

Outsource repetitive or challenging work to a large group of semi-organized individuals.

On-Demand Production

Produce items after an order has been received to avoid carrying costs of inventory.

Lean Production

Reduce waste and cost in your manufacturing process and other operations.

Logistics Systems

Manage the flow of goods, information and other resources between the point of origin and the point of use.

Strategic Design

Employ a purposeful approach that manifests itself consistently across offerings, brands, and experiences.

Intellectual Property

Protect an idea that has commercial value—such as a recipe or industrial process—with legal tools like patents.

User Generated

Put your users to work in creating and curating content that powers your offerings.

Predictive Analytics

Model past performance data and predict future outcomes to design and price offerings accordingly.

Product Performance

Superior Product

Develop an offering of exceptional design, quality, and/or experience.

Ease of Use

Make your product simple, intuitive and comfortable to use.

Engaging Functionality

Provide an unexpected or newsworthy experiential component that elevates the customer interaction.

Safety

Increase the customer's level of confidence and security.

Feature Aggregation

Combine existing features found across offerings into a single offering.

Added Functionality

Add new functionality to an existing offering.

Performance Simplification

Omit superfluous details, features, and interactions to reduce complexity.

Environmental Sensitivity

Provide offerings that do no harm—or relatively less harm—to the environment.

Conservation

Design your product so that customers can reduce their use of energy or materials.

Customization

Enable altering of the product or service to suit individual requirements or specifications.

Focus

Design an offering specifically for a particular audience at the expense of others.

Styling

Impart a style, fashion or image.

Product System

Complements

Sell additional related or ancillary products or services to a customer.

Extensions/Plug-ins

Allow first- or third-party additions that add functionality.

Product Bundling

Offer several products for sale as one combined product.

Modular Systems

Provide a set of individual components that can be used independently, but gain utility when combined.

Product/Service Platforms

Develop systems that connect with other, partner products and services to create a holistic offering.

Integrated Offering

Combine otherwise discrete components into a complete experience.

Service

Try Before You Buy

Let customers test and experience an offering before investing in it.

Guarantee

Remove customer risk of lost money or time stemming from product failure or purchase error.

Loyalty Programs

Provide benefits and/or discounts to frequent and high-value customers.

Added Value

Include an additional service/function as part of the base price.

Concierge

Provide premium service by taking on tasks for which customers don't have time.

Total Experience Management

Provide thoughtful, holistic management of the consumer experience across an offering's lifecycle.

Supplementary Service

Offer ancillary services that fit with your offering.

Superior Service

Provide service(s) of higher quality, efficacy, or with a better experience than any competitor.

Personalized Service

Use the customer's own information to provide perfectly calibrated service.

User Communities/Support Systems

Provide a communal resource for product/service support, use and extension.

Lease or Loan

Let customers pay over time to lower upfront costs.

Self-Service

Provide users with control over activities that would otherwise require an intermediary to complete.

Channel

Diversification

Add and expand into new or different channels.

Flagship Store

Create a store to showcase quintessential brand and product attributes.

Go Direct

Skip traditional retail channels and connect directly with customers.

Non-Traditional Channels

Employ novel and relevant avenues to reach customers.

Pop-up Presence

Create a noteworthy but temporary environment to showcase and/or sell offerings.

Indirect Distribution

Use others as resellers who take ownership over delivering the offering to the final user.

Multi-Level Marketing

Sell bulk or packaged goods to an affiliated but independent sales force that turns around and sells it for you.

Cross-selling

Place products, services, or information that will enhance an experience in situations where customers are likely to want to access them.

On-Demand

Deliver goods in real-time whenever or wherever they are desired.

Context Specific

Offer timely access to goods that are appropriate for a specific location, occasion, or situation.

Experience Center

Create a space that encourages your customers to interact with your offerings—but purchase them through a different (and often lower-cost) channel.

Brand

Co-Branding

Combine brands to mutually reinforce key attributes or enhance the credibility of an offering.

Brand Leverage

"Lend" your credibility and allow others to use your name—thus extending your brand's reach.

Private Label

Provide goods made by others under your company's brand.

Brand Extension

Offer a new product or service under the umbrella of an existing brand.

Component Branding

Brand an integral component to make a final offering appear more valuable.

Transparency

Let customers see into your operations and participate with your brand and offerings.

Values Alignment

Make your brand stand for a big idea or a set of values and express them consistently in all aspects of your company.

Certification

Develop a brand or mark that signifies and ensures certain characteristics in third-party offerings.

Customer Engagement

Process Automation

Remove the burden of repetitive tasks from the user to simplify life and make new experiences seem magical.

Experience Simplification

Reduce complexity and focus on delivering specific experiences exceptionally well.

Curation

Use a distinct point of view to separate the proverbial wheat from the chaff—and in the process create a strong identity for yourself and your followers.

Experience Enabling

Extend the realm of what's possible to offer a previously improbable experience.

Mastery

Help customers to obtain great skill or deep knowledge of some activity or subject.

Autonomy and Authority

Grant users the power to use your offerings to shape their own experience.

Community and Belonging

Facilitate visceral connections to make people feel they are part of a group or movement.

Personalization

Alter a standard offering to allow the projection of the customer's identity.

Whimsy and Personality

Humanize your offering with small flourishes of on-brand, on-message ways of seeming alive.

Status and Recognition

Offer cues that infer meaning, allowing users—and those who interact with them—to develop and nurture aspects of their identity.



ENGINEERS AND ENVIRONMENTAL FAILURES



Mechanical Engineering

- Chlorofluorocarbons (CFCs), invented in the 1930s, provided a safe alternative to ammonia and sulphur dioxide
- It enabled widespread home and commercial refrigeration and air conditioning HVAC systems
- R12 exhibited excellent energy efficiency and performance properties and eliminated toxicity, flammability, and corrosion concerns of the other chemicals



Mechanical Engineering

But, the delayed effect was stratospheric ozone depletion due to CFCs leaking from refrigerators, air conditioners, and similar technology operations.

Civil Engineering

- The dam was built between 1924 and 1926 in the town of Francis in Los Angeles
- On the evening of March 12, 1928, the dam collapsed
- The sudden release of water caused flooding and destruction of 1000 homes
- The collapse resulted in the death of about 430 people





Civil Engineering

- William Mulholland, the chief engineer, had no formal training
- The height of the dam was increased by 10 feet without changing the foundation design
- Poor site analysis and geological understanding



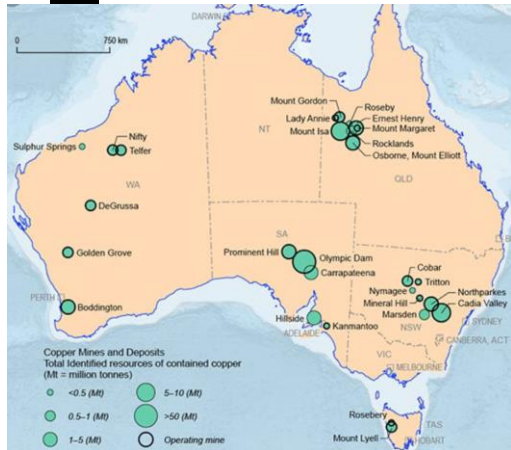
Chemical Engineering

- In 2003, Port Kembla Copper shut down its operations.
- Wide Ranging Health Problems in the community resulted in over 25 deaths over a 100 year period due to lead, arsenic, acid rain from SO_2 emissions.
- Three deaths in the workplace and numerous explosions

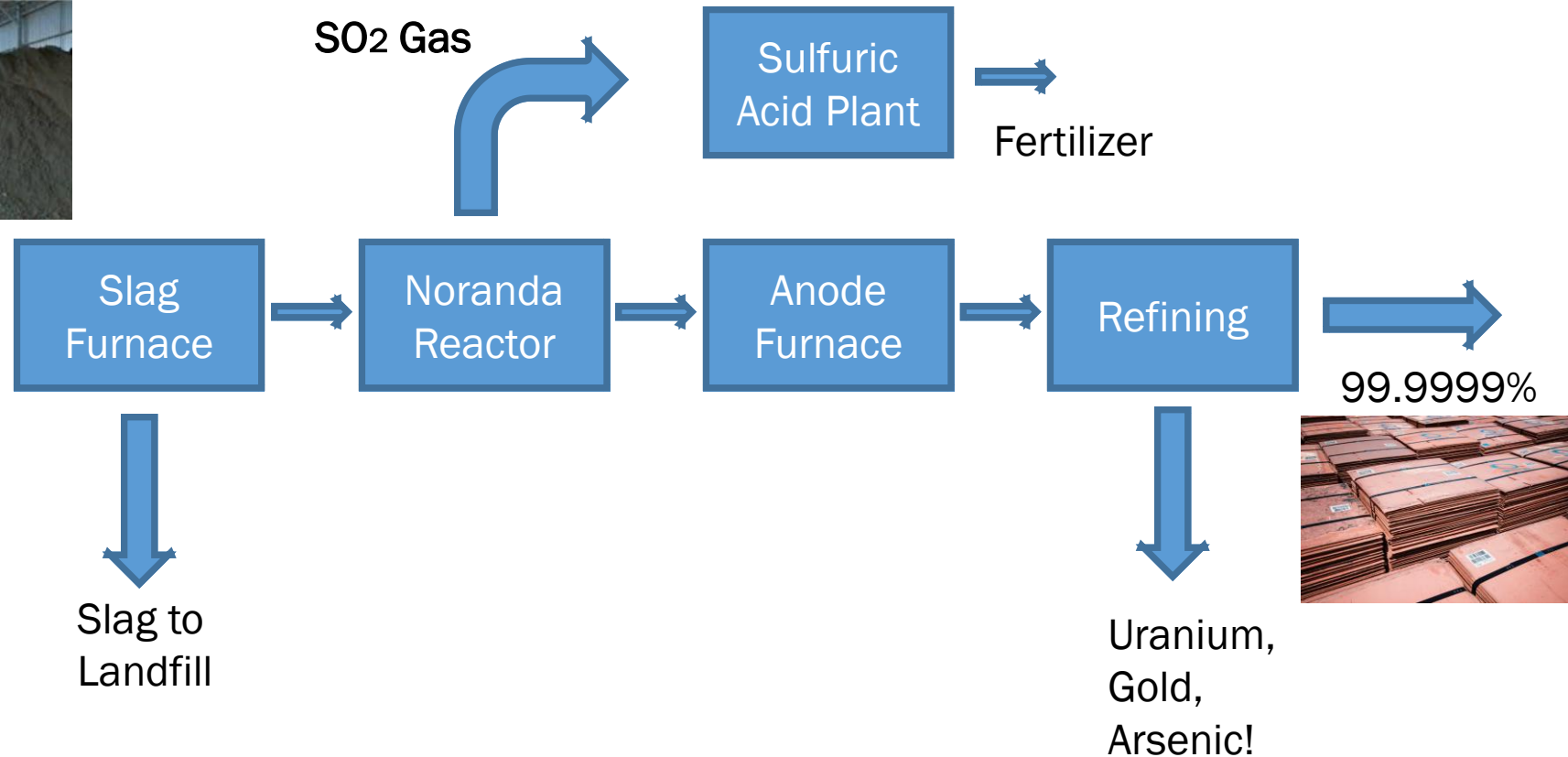
Port Kembla Copper, NSW Australia



Process to Produce Copper



30-40% Cu
Ore





Chemical Engineering

- PKC was an important part of the local industry but was located too close to urban population.
- Operated from 1908 to 2003
- The plant was located next to a primary school. The health impacts were well documented in children.
- Despite the plant being fitted with emission reduction technology, the Japanese operators withdrew from Australia.



Summary

- The course focuses on projects and on working in teams but with clear individual contributions
- 3 Clauses in the Code of Ethics relate to environmental matters
- Engineers need foresight to realise the impacts of their work prior to them happening
- We need systematic theory and tools to evaluate sustainability, impact and resilience of our engineering work